

Java OOPs Assignment Questions: Object, Class, and Constructor

Basic Level

1. Define a class `Student` with attributes `rollNumber`, `name`, and `marks`. Create an object of the class and print its data.
2. Write a Java program to create a `Book` class and instantiate two book objects using the class.
3. Create a class `Car` with fields `model`, `year`, and `price`. Instantiate and initialize an object using a constructor.
4. What is the role of the `new` keyword in object creation? Demonstrate with a code snippet using a `Person` class.
5. Differentiate between object and class in Java with a practical example.

Intermediate Level

6. Write a class `Employee` with two constructors - one default and one parameterized. Instantiate objects using both constructors.
7. Create a class `Circle` with a constructor that calculates and displays the area of the circle. Accept radius as input.
8. Create a `Rectangle` class with overloaded constructors for:
 - No parameters
 - One parameter (square)
 - Two parameters (length and breadth)
9. Define a `BankAccount` class with constructor initialization and a method to display balance. Write a program to demonstrate object behavior.
10. Explain what happens when a constructor is not explicitly defined in a class. Demonstrate with code.

Advanced Level

11. Create a class `Laptop` with a static counter to keep track of the number of objects created. Use constructor logic to increment the counter.
12. Develop a class `Product` and use constructor chaining to initialize fields. Demonstrate using `this()` keyword.
13. What is constructor overloading? Write a program to illustrate this concept using a `Student` class with

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different initialization options.

14. Implement a class `Movie` with private fields and use constructors along with getter methods to initialize and access data.

15. Demonstrate the use of a copy constructor in Java with a class `Book`.